Lecture 10
Serre Vanishing

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1 Review: Ample and Very Ample Line Bundles

Throughout these notes, all schemes are assumed to be noetherian unless stated otherwise.

1.1 Twisting sheaves

Let $\mathbb{P}^r_A = \operatorname{Proj} A[x_0, \dots, x_r]$. The twisting sheaf on $\mathbb{P}^r_A$ is defined by $\mathcal{O}_{\mathbb{P}^r_A}(1) = (A[x_0, \dots, x_r](1))^\sim$. For every integer n, we write $\mathcal{O}_{\mathbb{P}^r_A}(n) = \mathcal{O}_{\mathbb{P}^r_A}(1)^{\otimes n}$, where negative tensor powers are understood as tensor powers of the dual line bundle.

If $\mathcal{F}$ is an $\mathcal{O}_{\mathbb{P}^r_A}$-module, we define its twist by $\mathcal{F}(n) = \mathcal{F} \otimes \mathcal{O}_{\mathbb{P}^r_A}(n)$.

More generally, if X is a projective A-scheme equipped with a very ample invertible sheaf $\mathcal{O}_X(1)$, we define

$$\mathcal{O}_X(n) = \mathcal{O}_X(1)^{\otimes n}, \qquad \mathcal{F}(n) = \mathcal{F} \otimes \mathcal{O}_X(n).$$

1.2 Very ample line bundles

Definition 1.1. Let $f : X \to \operatorname{Spec} A$ be a scheme over A, and let $\mathcal{L}$ be an invertible sheaf on X.

We say that $\mathcal{L}$ is *very ample over* $\operatorname{Spec} A$ if there exists an immersion $i : X \hookrightarrow \mathbb{P}^r_A$ for some r such that $\mathcal{L} \cong i^*\mathcal{O}_{\mathbb{P}^r_A}(1)$.

When X is projective over A, the immersion above is automatically a closed immersion. Indeed, $X \to \operatorname{Spec} A$ is proper, while $\mathbb{P}^r_A \to \operatorname{Spec} A$ is separated. Hence the graph of i is a closed immersion, so i is proper. Since every proper immersion is a closed immersion, i is closed.

A fundamental characterization of projective morphisms is the following.

Theorem 1.2. *Suppose A is noetherian. A scheme X is projective over* $\operatorname{Spec} A$ *if and only if it is proper over* $\operatorname{Spec} A$ *and admits a very ample invertible sheaf relative to* $\operatorname{Spec} A$.

1.3 Ample line bundles

Definition 1.3. Let X be a noetherian scheme and let $\mathcal{L}$ be an invertible sheaf on X.

We say that $\mathcal{L}$ is *ample* if for every coherent sheaf $\mathcal{F}$ on X, there exists an integer n_0 such that for every $n \geq n_0$, the sheaf $\mathcal{F} \otimes \mathcal{L}^{\otimes n}$ is generated by finitely many global sections.

Unlike very ampleness, ampleness is an intrinsic property of the line bundle and does not depend on a morphism to a base scheme.

The following facts will be used later without proof.

Proposition 1.4. *Let $\mathcal{L}$ be an invertible sheaf on a noetherian scheme X.*

(1) Every very ample line bundle is ample.

(2) $\mathcal{L}$ is ample if and only if $\mathcal{L}^{\otimes m}$ is ample for every positive integer m.

(3) If X is proper over a noetherian ring A, then $\mathcal{L}$ is ample $\iff \mathcal{L}^{\otimes m}$ is very ample over $\operatorname{Spec} A$ *for some positive integer m.*

2 Serre's Global Generation Theorem

The following theorem is the main geometric ingredient in the proof of Serre's vanishing theorem.

Theorem 2.1 (Serre)**.** *Let X be a projective scheme over a noetherian ring A, and let $\mathcal{O}_X(1)$ be a very ample invertible sheaf on X relative to* $\operatorname{Spec} A$*. If $\mathcal{F}$ is a coherent sheaf on X, then there exists an integer n_0 such that for every $n \geq n_0$, the twisted sheaf $\mathcal{F}(n)$ is generated by finitely many global sections.*

Proof. Choose a closed immersion $i : X \hookrightarrow \mathbb{P}^r_A$ defining $\mathcal{O}_X(1)$, so that $\mathcal{O}_X(1) \cong i^*\mathcal{O}_{\mathbb{P}^r_A}(1)$. Since i is finite, $i_*\mathcal{F}$ is coherent on $\mathbb{P}^r_A$, and $H^0(X, \mathcal{F}(n)) \cong H^0(\mathbb{P}^r_A, i_*\mathcal{F}(n))$. Hence it suffices to prove the theorem for $X = \mathbb{P}^r_A$.

Cover $\mathbb{P}^r_A$ by the standard affine open subsets $D_+(x_0), \ldots, D_+(x_r)$. For each i, the restriction $\mathcal{F}|_{D_+(x_i)}$ is associated to a finitely generated module over $A\left[\frac{x_0}{x_i}, \ldots, \widehat{\frac{x_i}{x_i}}, \ldots, \frac{x_r}{x_i}\right]$. Choose finitely many generators on every affine chart. By the extension property of twisted sheaves, after twisting by a sufficiently large integer n, each local generator extends to a global section of $\mathcal{F}(n)$.

Since multiplication by x_i^n identifies the restriction of $\mathcal{F}$ with that of $\mathcal{F}(n)$ over $D_+(x_i)$, these global sections generate $\mathcal{F}(n)$ on every affine chart. Therefore they generate $\mathcal{F}(n)$ globally. □

Corollary 2.2. *Let X be a projective scheme over a noetherian ring and let $\mathcal{F}$ be a coherent sheaf on X. Then there exist an integer n and a positive integer N together with a surjective morphism*

$$\mathcal{O}_X(-n)^{\oplus N} \twoheadrightarrow \mathcal{F}.$$

Consequently, every coherent sheaf is a quotient of a finite direct sum of twisted structure sheaves.

Proof. Choose n sufficiently large so that $\mathcal{F}(n)$ is generated by N global sections. These sections induce a surjective morphism

$$\mathcal{O}_X^{\oplus N} \twoheadrightarrow \mathcal{F}(n).$$

Tensoring with the invertible sheaf $\mathcal{O}_X(-n)$ preserves surjectivity and yields

$$\mathcal{O}_X(-n)^{\oplus N} \twoheadrightarrow \mathcal{F}.$$

The final statement follows immediately. □

3 Serre Vanishing

We now prove the main theorem of this lecture.

Theorem 3.1 (Serre Vanishing)**.** *Let X be a projective scheme over a noetherian ring A, let $\mathcal{O}_X(1)$ be a very ample invertible sheaf on X, and let $\mathcal{F}$ be a coherent sheaf on X.*

(a) *For every $i \geq 0$, the A-module $H^i(X, \mathcal{F})$ is finitely generated.*

(b) *There exists an integer n_0 such that $H^i(X, \mathcal{F}(n)) = 0$ for every $i > 0$ and every $n \geq n_0$.*

Proof. Choose a closed immersion $i : X \hookrightarrow \mathbb{P}^r_A$ such that $\mathcal{O}_X(1) \cong i^*\mathcal{O}_{\mathbb{P}^r_A}(1)$. Since i is finite, $i_*\mathcal{F}$ is coherent on $\mathbb{P}^r_A$, and $H^i(X, \mathcal{F}) \cong H^i(\mathbb{P}^r_A, i_*\mathcal{F})$ for every i. Hence it is enough to prove the theorem for $X = \mathbb{P}^r_A$.

For sheaves of the form $\mathcal{O}(q)$, both assertions follow immediately from the explicit computation of $H^i(\mathbb{P}^r_A, \mathcal{O}(q))$. The same therefore holds for every finite direct sum of twisted structure sheaves.

For an arbitrary coherent sheaf $\mathcal{F}$, choose a short exact sequence

$$0 \longrightarrow \mathcal{R} \longrightarrow \mathcal{E} \longrightarrow \mathcal{F} \longrightarrow 0,$$

where $\mathcal{E} = \mathcal{O}(-n)^{\oplus N}$ for suitable integers n and N.

Since X is noetherian, the category of coherent sheaves on X is abelian. Therefore the kernel $\mathcal{R}$ is again coherent.

We argue by descending induction on the cohomological degree.

First observe that $H^i(X, \mathcal{G}) = 0,\ i > r$, for every quasi-coherent sheaf $\mathcal{G}$.

Indeed, $\mathbb{P}^r_A$ is covered by the $r + 1$ standard affine open subsets $D_+(x_0), \ldots, D_+(x_r)$. Since every finite intersection of these affine opens is again affine, Čech cohomology with respect to this covering computes sheaf cohomology. The corresponding Čech complex has length at most r, and therefore its cohomology vanishes in all degrees greater than r.

Assume that assertions (a) and (b) have already been established for cohomological degree $i + 1$.

The long exact cohomology sequence gives

$$H^i(X, \mathcal{E}) \longrightarrow H^i(X, \mathcal{F}) \longrightarrow H^{i+1}(X, \mathcal{R}).$$

Since both outer terms are finitely generated A-modules, exactness implies that $H^i(X, \mathcal{F})$ is also a finitely generated A-module. After twisting by $\mathcal{O}(m)$, we obtain another exact sequence

$$0 \longrightarrow \mathcal{R}(m) \longrightarrow \mathcal{E}(m) \longrightarrow \mathcal{F}(m) \longrightarrow 0.$$

Hence

$$H^i(X, \mathcal{E}(m)) \longrightarrow H^i(X, \mathcal{F}(m)) \longrightarrow H^{i+1}(X, \mathcal{R}(m)).$$

For sufficiently large m, $H^i(X, \mathcal{E}(m)) = 0$, because $\mathcal{E}$ is a finite direct sum of twisted structure sheaves.

By the induction hypothesis, $H^{i+1}(X, \mathcal{R}(m)) = 0$ for all sufficiently large m.

Therefore $H^i(X, \mathcal{F}(m)) = 0$ for all sufficiently large m. This completes the induction and hence the proof. □

4 The Vanishing Criterion for Ampleness

Serre vanishing can be extended from very ample line bundles to arbitrary ample line bundles. We prove the implication needed here.

Proposition 4.1. *Let X be a proper scheme over a noetherian ring A, and let $\mathcal{L}$ be an invertible sheaf on X. Suppose that $\mathcal{L}$ is ample. Then, for every coherent sheaf $\mathcal{F}$ on X, there exists an integer n_0 such that $H^i\big(X, \mathcal{F} \otimes \mathcal{L}^{\otimes n}\big) = 0$ for every $i > 0$ and every $n \geq n_0$.*

Proof. Since $\mathcal{L}$ is ample and X is proper over A, there exists an integer $m > 0$ such that $\mathcal{L}^{\otimes m}$ is very ample relative to $\operatorname{Spec} A$.

For each integer j with $0 \leq j < m$, consider the coherent sheaf $\mathcal{F}_j = \mathcal{F} \otimes \mathcal{L}^{\otimes j}$. Applying Serre vanishing to the very ample line bundle $\mathcal{L}^{\otimes m}$ and the coherent sheaf $\mathcal{F}_j$, we obtain an integer q_j such that $H^i\left(X, \mathcal{F}_j \otimes \left(\mathcal{L}^{\otimes m}\right)^{\otimes q}\right) = 0$ for every $i > 0$ and every $q \geq q_j$.

Since there are only finitely many possible values of j, we may set $Q = \max\{q_0, \ldots, q_{m-1}\}$. Then, for every $0 \leq j < m$, every $i > 0$, and every $q \geq Q$, we have $H^i\left(X, \mathcal{F} \otimes \mathcal{L}^{\otimes(mq+j)}\right) = 0$.

Every sufficiently large integer n can be written uniquely in the form $n = mq+j,\ 0 \leq j < m$, with $q \geq Q$.

It follows that $H^i\big(X, \mathcal{F} \otimes \mathcal{L}^{\otimes n}\big) = 0$ for every $i > 0$ and every sufficiently large n. □

Remark 4.2. For a very ample line bundle $\mathcal{O}_X(1)$, the preceding proposition reduces to the usual form of Serre vanishing:

$$H^i(X, \mathcal{F}(n)) = 0 \qquad (i > 0,\ n \gg 0).$$

The passage from very ample to ample line bundles is achieved by choosing a positive power of $\mathcal{L}$ which is very ample and treating the finitely many residue classes modulo that power separately.

References

[1] R. Hartshorne, *Algebraic Geometry*, Graduate Texts in Mathematics, vol. 52, Springer-Verlag, 1977.